

Machine Learning: A Bird's Eye View

By

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Machine Learning

- **Herbert Alexander Simon:**
“Learning is any process by which a system improves performance from experience.”
- “Machine Learning is concerned with computer programs that automatically improve their performance through experience. “



Herbert Simon
[Turing Award](#) 1975
[Nobel Prize in Economics](#) 1978

Why “Learn”?

- Machine learning is programming computers to **optimize** a performance criterion using example data which can act like **past experience**.
- There is no need to “learn” to calculate payroll
- Learning is used when:
 - Human expertise does not exist (navigating on Mars),
 - Humans are unable to explain their expertise (speech recognition)
 - Solution changes in time (routing on a computer network)
 - Solution needs to be adapted to particular cases (user biometrics)

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What We Talk About When We Talk About “Learning”

- Learning general models from a data of particular examples
- Data is cheap and abundant (data warehouses, data marts); knowledge is expensive and scarce.
- Example in retail: Customer transactions to consumer behavior:

People who bought “Da Vinci Code” also bought “The Five People You Meet in Heaven” (www.amazon.com)
- Build a model that is *a good and useful approximation* to the data.

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What is Machine Learning?

Machine learning is a field of computer science that uses statistical techniques to give computer systems the ability to "learn" (i.e., progressively improve performance on a specific task) with data, without being explicitly programmed.

The name machine learning was coined in 1959 by Arthur Samuel. - Wikipedia

Traditional Programming

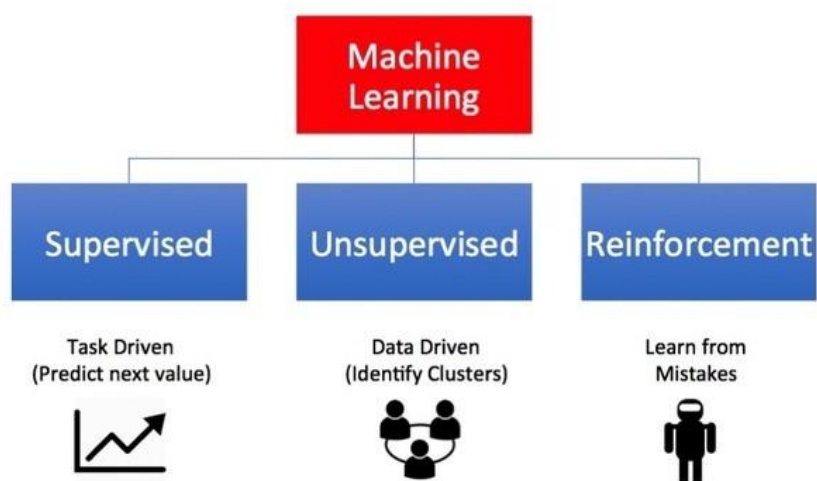


Machine Learning



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Types of Machine Learning



<https://towardsdatascience.com/what-are-the-types-of-machine-learning-e2b9e5d1756f>

Conventionally....

Machine Learning Problems

	<i>Supervised Learning</i>	<i>Unsupervised Learning</i>
<i>Discrete</i>	classification or categorization	clustering
<i>Continuous</i>	regression	dimensionality reduction

The data and the goal

- **Data:** A set of data records (also called examples, instances or cases) described by
 - *k* attributes/variables/features: $A_1, A_2, \dots, A_k$.
 - a class: Each example is labelled with a pre-defined class.
- **Goal:** To learn a classification model f from the data that can be used to predict the classes of new (future, or test) cases/instances.

An example: data (loan application)

Approved or not

ID	Age	Has_Job	Own_House	Credit_Rating	Class
1	young	false	false	fair	No
2	young	false	false	good	No
3	young	true	false	good	Yes
4	young	true	true	fair	Yes
5	young	false	false	fair	No
6	middle	false	false	fair	No
7	middle	false	false	good	No
8	middle	true	true	good	Yes
9	middle	false	true	excellent	Yes
10	middle	false	true	excellent	Yes
11	old	false	true	excellent	Yes
12	old	false	true	good	Yes
13	old	true	false	good	Yes
14	old	true	false	excellent	Yes
15	old	false	false	fair	No

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An example: the learning task

- Learn a classification model from the data
- Use the model to classify future loan applications into
 - Yes (approved) and
 - No (not approved)
- What is the class for following case/instance?

Age	Has_Job	Own_house	Credit-Rating	Class
young	false	false	good	?

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The machine learning framework

- Apply a prediction function to a feature representation of the image to get the desired output:

$$f(\text{apple image}) = \text{"apple"}$$

$$f(\text{tomato image}) = \text{"tomato"}$$

$$f(\text{cow image}) = \text{"cow"}$$

■ Slide credit: L. Lazebnik

The machine learning framework

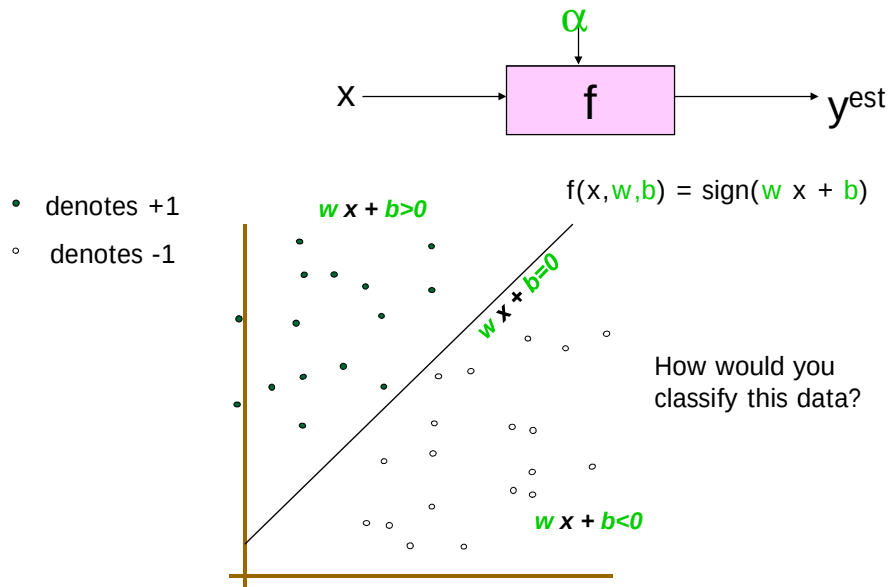
$$y = f(x)$$

output prediction function Image feature

- **Training:** given a *training set* of labeled examples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, estimate the prediction function f by minimizing the prediction error on the training set
- **Testing:** apply f to a never before seen *test example* $\mathbf{x}$ and output the predicted value $y = f(\mathbf{x})$

■ Slide credit: L. Lazebnik

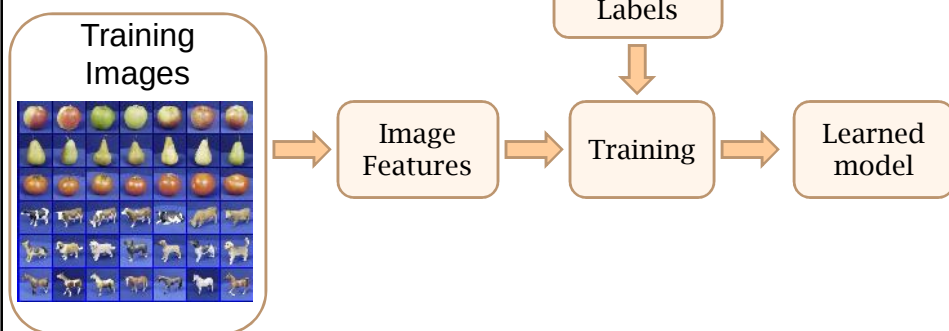
Linear Classifiers: The Simple Models!



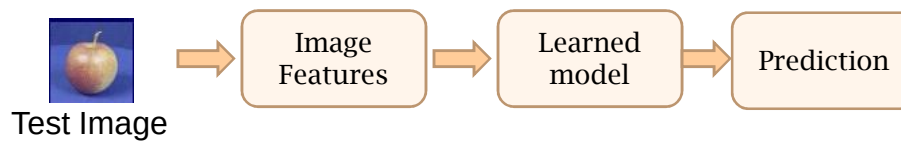
Steps

Slide credit: D. Hoiem and L. Lazebnik

Training



Testing



Is ML really so hard?

Input: X

Output: Y



Label "motorcycle"

Slide courtesy: Dr. Mingxuan Sun, LSU

Why is it hard?

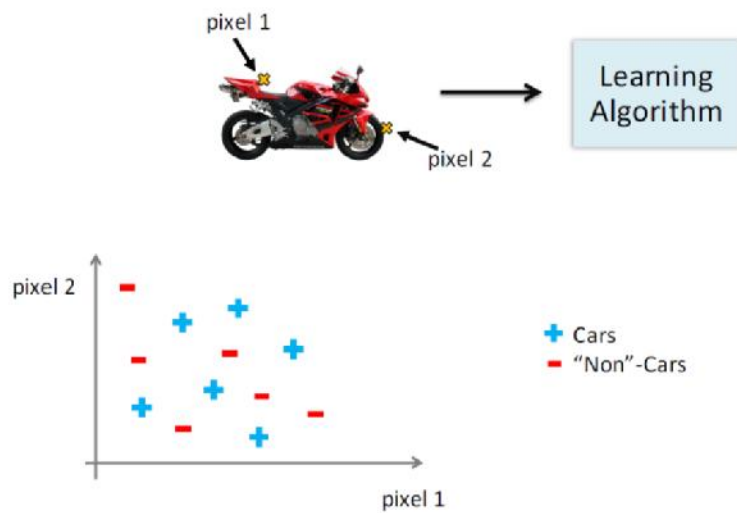
You see this



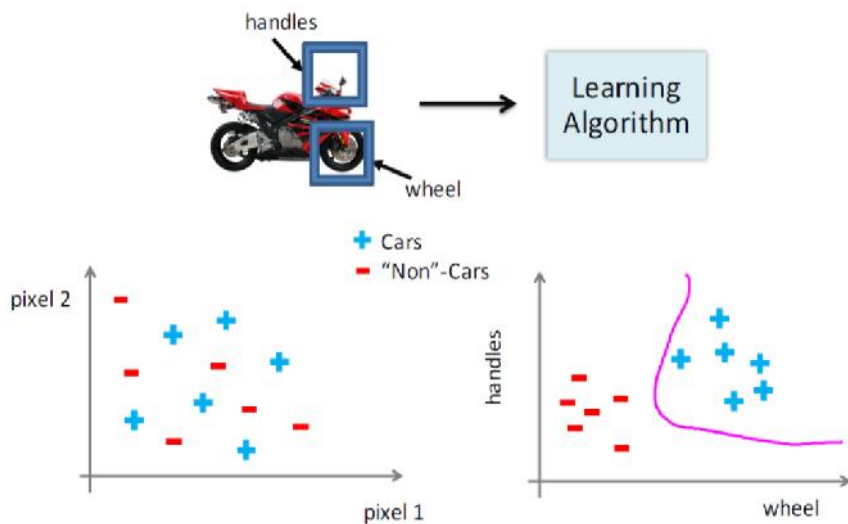
But the camera sees this:

194	210	201	212	199	213	215	195	170	150	102	209
180	189	190	221	209	205	191	167	147	115	129	163
114	126	140	188	176	165	152	140	170	106	78	88
87	103	115	154	143	142	149	153	173	101	57	57
102	112	106	131	122	139	152	147	120	04	50	66
94	95	79	104	105	124	129	113	107	07	69	67
50	71	69	90	09	92	90	95	09	09	76	67
41	56	68	99	63	45	60	62	50	75	75	65
20	43	69	75	56	41	51	73	55	70	63	44
50	50	57	69	75	75	73	74	53	68	59	37
72	59	53	66	84	92	84	74	57	72	63	42
67	61	50	65	75	78	76	73	59	75	69	50

Raw Image Representation



Better Feature Representation?



But still classification is not that easy....especially nowadays...

Labradoodle or fried chicken



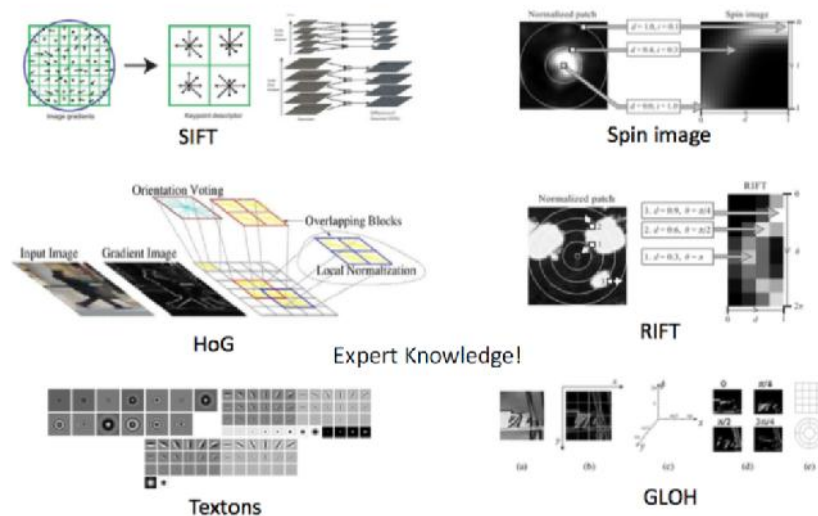
Barn owl or apple



Chihuahua or muffin



Perhaps a lot of it depends on proper feature representations: **Feature Engineering!**



So many classifiers over the years....

- k-nearest neighbor
- SVM
- Decision Trees
- Neural networks
- Naïve Bayes
- Bayesian network
- Logistic regression
- Randomized Forests
- The Deep Learning Systems
- And so on.....

And then comes the No Free Lunch Theorem of ML.....

Generative vs. Discriminative Classifiers

Generative Models

- Represent both the data and the labels
- Often, makes use of conditional independence and priors
- Examples
 - Naïve Bayes classifier
 - Bayesian network
 - GANs
- Models of data may apply to future prediction problems

Discriminative Models

- Learn to directly predict the labels from the data
- Often, assume a simple boundary (e.g., linear)
- Examples
 - Logistic regression
 - SVM
 - Boosted decision trees
- Often easier to predict a label from the data than to model the data

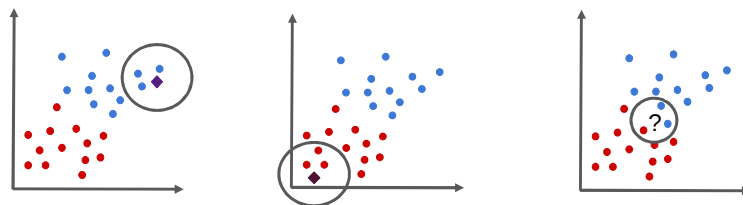
Slide credit: D. Hoiem

A very brief look into a few traditional classifiers

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The k -Nearest Neighbor Classifier

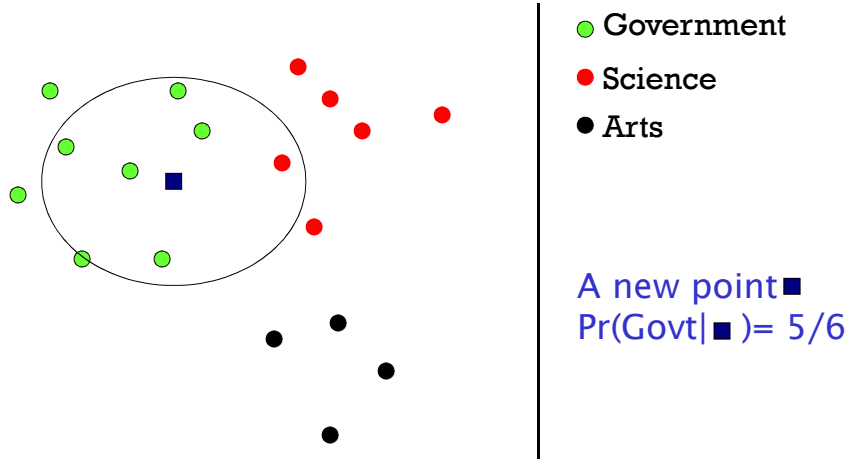
The k -Nearest Neighbor (kNN) classifier (Fix and Hodges 1958, Cover and Hart 1967) labels a test point y , by that class which has the majority number of representatives among the training set neighbors of y .



Reference:

- E. Fix and J.L. Hodges, Discriminatory analysis-nonparametric discrimination: consistency properties, Technical Report, California Univ Berkeley (1951)
- T. M. Cover and P. E. Hart, "Nearest neighbour pattern classification," IEEE Transactions on Information Theory, vol. 13, no. 1, pp. 21–27, 1967.

Example: k=6 (6NN)



kNN

- Parametric estimation technique
- Approximation Maximum Posterior (MAP) rule
 - Maximum relative function class
- Training set of example with known class labels
- If you want to classify a new example
- Find its **distance** to each of the training example
- Find **the k-nearest neighbors**
- Find the class which has the highest number among **these k-neighbor (winner)** and assign that class label

How to measure the distance?

- Euclidean Distance

$$D(x, y) = \sum_i (x^i - y^i)^2$$

- Minkowsky distance

$$D(x, y) = \left(\sum_i (x^i - y^i)^r \right)^{\frac{1}{r}}$$

- Manhattan distance

$$D(x, y) = \sum_i |x^i - y^i|$$

- Lamberra distance

$$D(x, y) = \sum_i \left| \frac{x^i - y^i}{x^i + y^i} \right|$$

- Chebychav distance
- Mahalanobis distance
- Correlation

How to measure k?

- K is too small → sensitive to noise
- K is large neighborhood may include points from other class

- Thumb rule:

- $k = \sqrt{N}$ or $k = N/10$

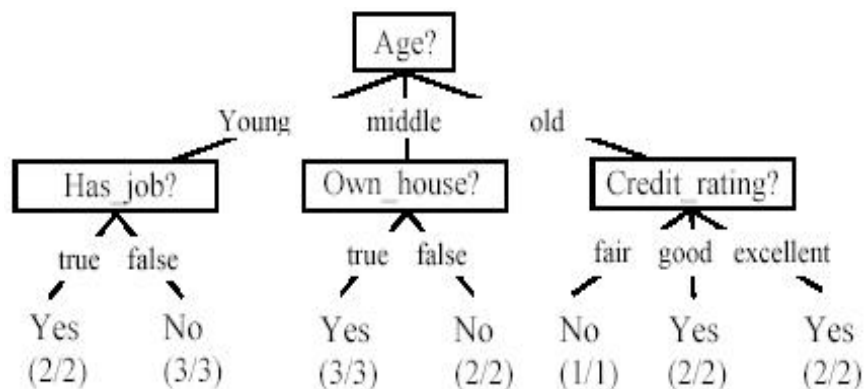
- Open problem in data mining

Time Complexity

- $O(kDN^2)$
- K is number of nbd
- D is the dimension of data
- N is training sample size
- $O(kn) \rightarrow$ for sorting to first k- distance
- $O(DN) \rightarrow$ to find N distance

The Decision Tree Classifier

Decision nodes and leaf nodes (classes)



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Decision Tree Classifiers....

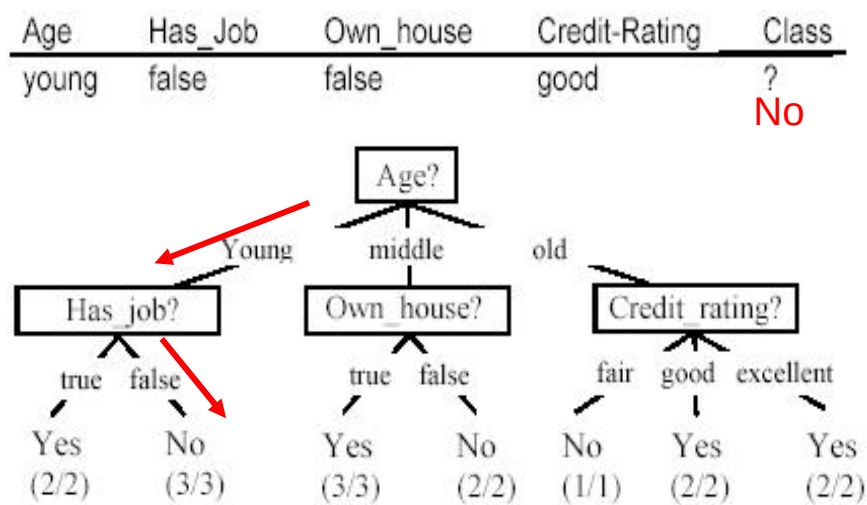
The Loan Data Reproduced...

Approved or not

ID	Age	Has_Job	Own_House	Credit_Rating	Class
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Use the decision tree



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Is the decision tree unique?

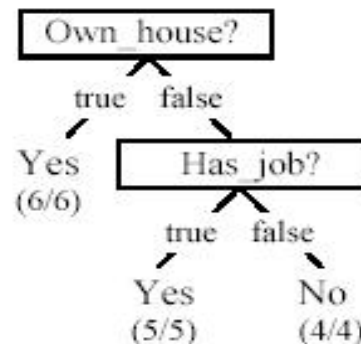
No. Here is a simpler tree.

We want **smaller tree** and **accurate tree**.

Easy to understand and perform better.

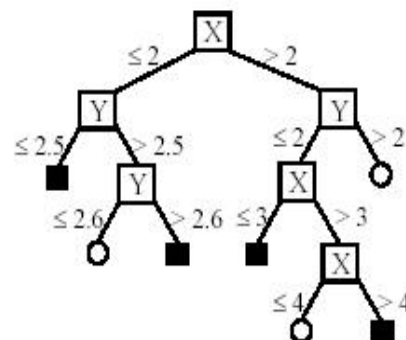
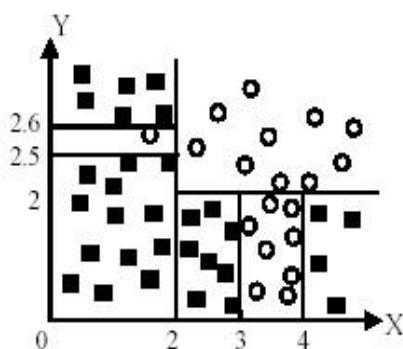
Finding the best tree is
NP-hard.

All current tree building
algorithms are heuristic
algorithms



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An example in a continuous space



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